

Tactical Analysis Using Graph Neural Networks in Professional Soccer

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Abstract—This project investigated the application of Spatiotemporal Graph Neural Networks (STGNNs) for tactical analysis in professional soccer. The exploration aimed to address the limitations of traditional, event-based metrics by modeling the sport as a relational system where players function as interconnected nodes within a dynamic graph. Utilizing the StatsBomb 360 dataset, a model was developed to perform team-level formation recognition and individual tactical evaluation. Technical explorations focused on the impact of aggregation functions and temporal windowing, specifically identifying that sum-based aggregation is superior for quantifying spatial density in team structures.

The resulting architecture achieved a validation accuracy of 91.14%, demonstrating high proficiency in identifying complex geometries such as the 3-5-2 formation. Beyond the technical implementation, this project highlighted the inherent discrepancy between static coaching labels and the fluid reality of match dynamics. The exploration fostered a deeper understanding of deep learning workflows, data storytelling, and the challenges of data sparsity in professional tracking. These experiences confirmed a professional interest in machine learning engineering and provided a robust foundation for continued graduate research in applied data science.

I. INTRODUCTION

Soccer is a relational system. At any moment during a match, every player’s position, movement, and decision is shaped by the positions and movements of every other player on the pitch. Traditional analytics metrics like expected goals (xG) [1], pass completion rates, or shot maps, capture individual events well but fail to represent this relational structure. A through ball is not just a pass; it is a spatial interaction between a passer, a runner, a defensive line, and the lane between them.

Graph Neural Networks (GNNs) are well suited to this problem. Unlike conventional neural networks, which assume input data is organized in fixed-size vectors or grids, GNNs operate on graph-structured data where entities and their relationships are jointly encoded [7]. A soccer team is naturally represented as a graph: players are nodes, spatial, tactical, and statistical relationships are edges, and as players move, the graph evolves.

This paper examines GNNs applied to two tasks in professional soccer: formation recognition, framed as supervised classification over team-level graph embeddings, and player tactical analysis, encompassing counterattack success prediction and the identification of pivotal players whose contributions are invisible to event-based metrics and statistics. Together these two lenses illustrate how a single architectural framework can operate at both the team level and the individual node level in professional soccer.

II. GRAPH NEURAL NETWORKS

A. Representing Soccer as a Graph

A graph is a structure of nodes and edges, where nodes represent entities and edges represent relationships. In soccer, each player on a team is a node. Each node carries a feature vector describing the player's current state, most commonly spatial coordinates (x, y) , velocity, and role indicators such as whether the player is a goalkeeper or outfield player with flexibility to add more.

The choice of edge construction strategy plays a large role in what relationships the model can learn. Three strategies are common:

- **Euclidean threshold:** connects players within a fixed distance (e.g., 10 meters). Simple but sensitive to player density.
- **Delaunay triangulation:** connects each player to their natural spatial neighbors by partitioning the pitch into non-overlapping triangles. Sparser and more structurally meaningful than raw proximity [12], [14].
- **Voronoi adjacency:** partitions the pitch into regions of territorial influence and connects players whose regions share a boundary.

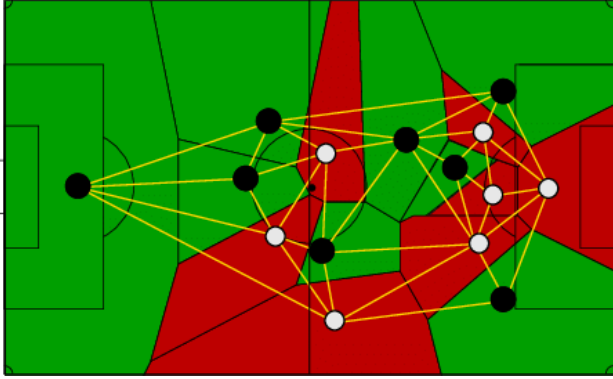


Fig. 1. Voronoi adjacency and Delaunay Triangulation during a soccer situation.[8] Displays the Voronoi diagram with green and red shapes displaying each player's territorial region on the pitch. While Delaunay triangulation is shown with yellow lines visualizing the connected nature of a soccer match.

B. Message Passing

At the core of every GNN is message passing. At each layer, every node aggregates information from its neighbors and updates its own representation. After k layers, each node's embedding reflects the structure of its k -hop neighborhood [10].

Formally, the update at layer l for node v is:

$$h_v^{(l)} = \text{UPDATE}\left(h_v^{(l-1)}, \text{AGG}\left(\left\{w_{uv} \cdot h_u^{(l-1)} : u \in \mathcal{N}(v)\right\}\right)\right)$$

where $h_v^{(l)}$ is the embedding of node v at layer l , $\mathcal{N}(v)$ is the neighbor set of v , w_{uv} is the edge weight between u and v , AGG is a permutation-invariant aggregation function, and UPDATE combines the prior state with the aggregated neighborhood. The choice of AGG is not a trivial implementation detail. It encodes an assumption about what relational

information the model should preserve. In other words, it decides what insights your model looks for.

C. Aggregation Functions

Sum aggregation accumulates the weighted sum of all neighbor embeddings. The result grows with both the number of neighbors and their edge weights, making it sensitive to neighborhood size. A central midfielder connected to six teammates accumulates a richer signal than a wide forward with two connections. This sensitivity makes sum well suited for formation recognition, where a formation's structural signature is encoded in the collective connectivity pattern of the whole team [12].

Mean aggregation normalizes by neighbor count, discarding neighborhood size and retaining only the average character of the neighborhood. This is more robust to player density variation. For example, in a high press, players compress into a small area and sum aggregation would disproportionately amplify central nodes. However, this comes with a meaningful tradeoff: a player with six connections occupies a fundamentally different structural role than one with two, and mean aggregation loses that signal [9].

Max aggregation retains only the most prominent signal from the neighborhood. This winner-takes-all behavior suits tasks where one dominant interaction should drive the node update, a tight marker or an overlapping fullback, but it cannot represent the cumulative relational context of a player's full neighborhood, making it poorly suited for formation recognition [9].

Attention aggregation, used in Graph Attention Networks (GATs), replaces fixed edge weights with learned coefficients computed dynamically from node features [13]. Two players at identical spatial distances may receive very different attention weights if their feature vectors suggest different tactical relevance. This makes attention the most expressive of the four strategies and the natural choice for player tactical analysis, where the goal is identifying which specific relationships drive individual contributions [8], [9].

Table I summarizes these tradeoffs.

TABLE I
AGGREGATION FUNCTION COMPARISON: CATEGORIZES GNN AGGREGATION TECHNIQUES BY THEIR ABILITY TO PRESERVE NEIGHBORHOOD SIZE AND THEIR PRIMARY TACTICAL USE CASE. SUM IS IDENTIFIED AS THE OPTIMAL CHOICE FOR FORMATION RECOGNITION DUE TO ITS SENSITIVITY TO PLAYER DENSITY.

Function	Preserves Size	Learned	Best For
Sum	Yes	No	Formation recognition
Mean	No	No	Density-robust tasks
Max	No	No	Dominant interactions
Attention	No	Yes	Tactical analysis

III. SPATIOTEMPORAL GRAPH NEURAL NETWORKS

A static graph captures the team at a single instant. Soccer is a continuous temporal process: formations shift, transitions occur, and tactical states evolve across a match. A **Spatiotemporal Graph Neural Network (STGNN)** extends message passing across time by operating on a sequence of graphs

$\{G_1, G_2, \dots, G_T\}$, where each $G_t = (V, E_t)$ encodes the team's relational structure at time step t . The node set V is fixed, the same eleven players appear in every graph, but the edge set E_t is recomputed at each step as players move.

A. Spatial and Temporal Message Passing

The STGNN performs message passing in two stages. The spatial stage applies standard message passing within each G_t , producing node embeddings $h_v^{(L,t)}$ that encode each player's local relational context at that moment. The temporal stage aggregates these embeddings across time:

$$H_v^{(t)} = \text{TEMPORAL}\left(h_v^{(L,t)}, h_v^{(L,t-1)}, \dots, h_v^{(L,t-\tau)}\right) \quad (2)$$

where τ is the temporal window length and TEMPORAL is typically a recurrent mechanism such as a Long Short-Term Memory or Gated Recurrent Unit, or a temporal convolution [9], [11]. The result $H_v^{(t)}$ reflects both the player's current relational context and how that context has evolved over the preceding τ steps.

B. Architectural Limitations

Two design tensions constrain STGNNs for soccer. The first is the temporal window tradeoff. A short τ keeps the model sensitive to rapid tactical changes: a counterattack developing over three seconds, a defensive line stepping up to catch an attacker offside. A long τ provides more context about stable structural states, useful for formation recognition. But if a formation change occurs midway through a long window, the model's embedding reflects a mixture of the old and new structures, degrading classification precisely at the moments of tactical transition that are most analytically interesting [11]. Formation recognition benefits from longer windows; counterattack prediction requires shorter ones. No single value of τ optimally serves both [9].

The second tension is oversmoothing. As the number of message-passing layers L increases, each node aggregates from an increasingly large neighborhood. In a fully connected team graph, enough layers means every node is aggregating from every other node, and embeddings converge toward a common representation [10]. For formation recognition this is a direct threat: the classifier depends on meaningful differences between node embeddings—a goalkeeper's embedding should look different from a striker's. Too few layers and the model only sees local structure; too many and it loses the role distinctions that define a formation. Current mitigations include residual connections, normalization, and limiting network depth, but determining the right depth for a given application remains open [14], [9].

IV. FORMATION RECOGNITION

A. The Classification Problem

A soccer formation describes how a team's players are distributed across the pitch, expressed as a code such as 4-3-3 or 4-2-3-1. Formations are among the most fundamental tactical decisions a coach makes, shaping how a team defends,

transitions, and attacks. But the relationship between a formation label and the actual spatial arrangement of players at any given moment is far from rigid. Players drift, press, track runners, and rotate continuously, meaning the same formation can produce dramatically different spatial configurations depending on the phase of play [3], [4].

Early approaches assigned players to positional roles using average positions over a match segment, then matched those averages to a predefined template. Bialkowski et al. demonstrated that role-based positional clustering over large tracking datasets could identify dominant formations with reasonable accuracy, but the method collapses the continuous temporal evolution of team shape into a single static summary [6]. Elastic Formation and Position Identification improves on this by using linear sum assignment to optimally match players to formation templates frame-by-frame [4], but template-matching methods are fundamentally constrained by their predefined template set and encode no understanding of why a spatial arrangement corresponds to a particular tactical structure. Bauer et al. trained a convolutional neural network directly on positional snapshots, achieving an average F1-score of 0.76 across formation classes in Bundesliga tracking data [3], but convolutional models treat player positions as entries in a fixed spatial grid, discarding the relational structure between players—the very structure that defines a formation.

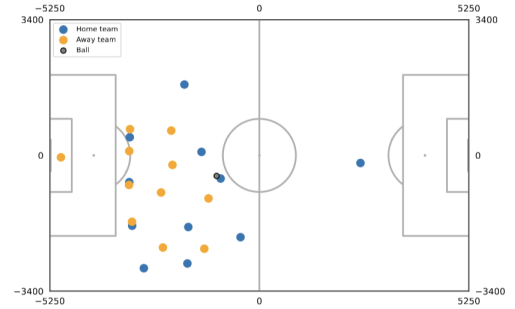


Fig. 2. Example frame from [14] — Lech Poznań vs Zagłębie Lubin, season 2023/24. The 2024 meeting between Poznań and Lubin where each player on the pitch and the ball is plotted with a colored dot. This is a visual example of what match data would look like before it is turned into a graph for a GNN.

B. The GNN Approach

Graph Neural Networks address formation recognition by operating directly on the relational structure of the team graph. Rather than asking *where* the players are, the model asks *how* they are connected, and learns that certain connectivity patterns correspond to certain formations regardless of where on the pitch those patterns occur [2].

Formation recognition is framed as a supervised classification task. The model is trained on labeled team graph snapshots and learns to map graph structure to formation class. The key insight is that a 4-3-3 produces a characteristic relational signature: a dense defensive block of four, a triangulated midfield trio, two wide attacking nodes flanking a central

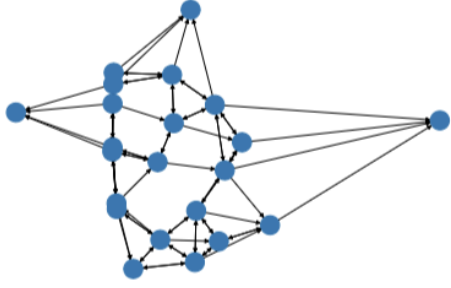


Fig. 3. Graph constructed from frame in 2 [14]. Figure 2 turned into a graph. This represents a single temporal frame ($\tau = 1$), precursor to a STGNN.

forward. The GNN learns to recognize this signature even as individual players move continuously within their roles.

Swiątek et al. demonstrate this using an overcomplete graph autoencoder with EdgeConv layers trained on high-frequency tracking data from the Polish Ekstraklasa [14]. EdgeConv constructs edges dynamically at each layer based on the current feature space rather than fixed spatial proximity, allowing the model to discover tactically relevant connections that static edge strategies might miss. Trained with a contrastive learning objective (pulling embeddings of similar tactical states together and pushing dissimilar states apart) the model clusters match frames by formation and identifies moments of rapid structural transition without requiring predefined formation templates [14].

V. PLAYER TACTICAL ANALYSIS

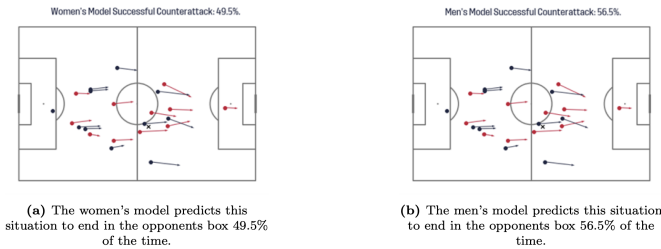


Fig. 4. Example of difference between men and women in Bekkers and Sahasrabudhe's findings [5]. They identified significant variances in directional momentum and spatial positioning during successful counterattacks.

A. Counterattack Success Prediction

A counterattack occurs when a team transitions rapidly from a defensive to an attacking state after regaining possession, exploiting the disorganization of the opposing defense before it can recover its shape. From a graph perspective, counterattacks are not isolated individual actions, they represent a team-wide reconfiguration of the graph. Spatial relationships, edge weights, and the directional momentum of nodes all shift simultaneously as the transition unfolds [5].

Bekkers and Sahasrabudhe model counterattack success as a graph-level classification problem using GNNs trained on synchronized spatiotemporal tracking data and event logs drawn

from MLS, NWSL, and international soccer [5]. Each frame of a counterattack sequence is a graph, with players as nodes carrying feature vectors encoding spatial coordinates, velocity, angle to goal, and angle to the ball. Permutation feature importance analysis identifies endline-to-endline speed, angle to goal, angle to the ball, and sideline-to-sideline speed as the node features with the highest impact on model performance, reflecting the intuition that counterattack success depends on directional momentum and spatial positioning relative to the goal, not just proximity between players.

A particularly notable finding is that gender-specific models, trained separately on men's and women's match data, outperform identical gender-ambiguous models trained on combined data. This reveals that the relational patterns underlying successful counterattacks differ meaningfully between competitions. Counterattacks exhibit competition-specific relational patterns that are lost when men's and women's data are combined into a single model [4].

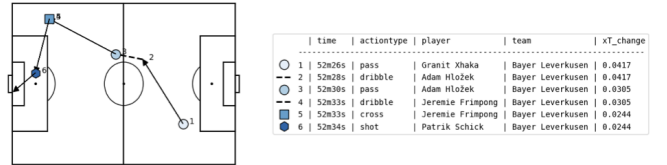


Fig. 5. Example of Jiang's model with xT [8]. For example, something like a long pass by Granit Xhaka is not something that would show up on a stat sheet. But with Jiang's xT model its value in the buildup to the goal is shown in the data

B. Identifying Pivotal Players

Traditional soccer metrics are biased toward attacking contributions. Expected goals rewards players who take shots; pass completion rates favor players who take low-risk passes. Both evaluate players by the discrete events they are directly involved in, ignoring the relational context that makes those events possible. A defensive midfielder who consistently intercepts passing lanes, connects defensive and attacking phases, and maintains structural balance in the team graph may contribute greatly to match outcomes while accumulating almost no credit in event-based statistics [8].

Jiang et al. address this with GoalNet, a GNN-based framework that assigns individual credit for changes in expected threat (xT) across both scoring and non-scoring actions [8]. GoalNet encodes each match sequence as an event-centric graph and enriches node embeddings with centrality measures, graph-theoretic quantities that capture how central a node is to the flow of information through the graph. A player who frequently appears on the shortest paths between other players, or whose removal would most disrupt the team's connectivity, receives a high centrality score regardless of whether they appear in the shot or goal log. Players widely recognized by tactical analysts as pivotal but undervalued by conventional metrics receive substantially higher credit scores under GoalNet than under xG-based systems [8].

Together, counterattack prediction and pivotal player identification illustrate how the same GNN framework operates at

two different scales. Counterattack prediction reads the team graph as a whole and predicts a collective outcome. Pivotal player identification reads the same graph at the node level and surfaces contributions that are invisible to event-based metrics. In both cases the analytical power comes from the same source: message passing encodes not just what players do, but how they relate to everyone around them.

VI. EXPLORATIONS

My exploration was driven by a desire to use a data structure to recognize the high-stakes, fluid environment of professional soccer analytics. I focused on the problem of tactical decipherability: using machine learning to interpret a match through the precision of spatiotemporal data, mirroring the intuition of a professional coaching staff.

A. From Static GNN to STGNN

I initially investigated whether a standard GNN could identify a team's shape from a single freeze frame of StatsBomb 360 tracking data. While it can make a prediction from a single moment, it lacked the memory to distinguish between a temporary attacking transition and a sustained defensive structure. This led to a technical pivot toward a STGNN architecture. By integrating a GRU to process sequences of fifteen consecutive frames ($\tau = 15$), the model gained the temporal dimension necessary to distinguish stable formations from transient positional states.

A key architectural decision was the use of sum-based aggregation in the spatial layers. As established in Table I, sum aggregation is sensitive to neighborhood size, allowing the model to quantify the total spatial density of specific tactical zones. A cluster of defenders grouped tightly in a defensive block accumulates a substantially larger aggregated signal than a solitary player in the same region, and that difference is precisely what formation recognition requires.

The technical outcome was a model achieving a final validation accuracy of 91.14%. Table II reports performance metrics.

TABLE II
MODEL PERFORMANCE BY FORMATION: METRICS ACROSS COMMON TACTICAL FORMATIONS. NOTE THE EXCEPTIONALLY HIGH PERFORMANCE IN IDENTIFYING THE 3-5-2 STRUCTURE, LIKELY DUE TO ITS UNIQUE SPATIAL SIGNATURE.

Formation	Precision	Recall	F1-Score
4-3-3	0.89	0.94	0.91
4-4-2	0.92	0.86	0.89
4-2-3-1	0.88	0.88	0.88
3-5-2	0.96	0.97	0.97
Overall	Validation Accuracy:		91.14%

B. The Latent Space: Formations Are Not Discrete

The t-distributed Stochastic Neighbor Embedding (t-SNE) visualization of the model's latent embeddings (Fig. 6) reveals something more interesting than the accuracy number alone. The 3-5-2 is the most isolated cluster, consistent with its geometrically distinct structure—five midfielders produce a

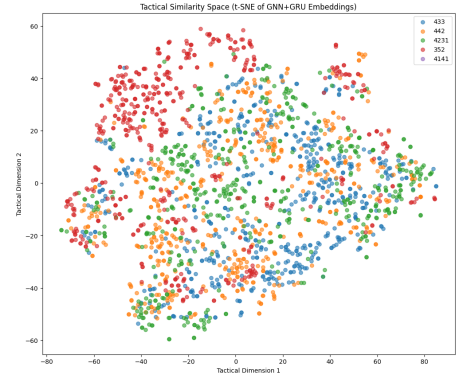


Fig. 6. t-SNE visualization of latent tactical clusters. This illustrates how the model develops an internal logic for formation similarity. The mixing of the clusters suggests the fluidity between formations mirroring what is seen in the game of professional soccer today.

connectivity signature that is genuinely separable from four-defender shapes. But all other formations cluster closely together, with substantial overlap at their boundaries.

This is not a failure of the model. It reflects the actual structure of professional soccer. Two teams both labeled 4-3-3 can play very differently depending on their defensive compactness and pressing triggers, while a 4-2-3-1 in possession may be spatially indistinguishable from a 4-3-3. The mixing in the latent space implies that formations are not discrete tactical identities—they are regions in a continuous space of spatial relationships, and many teams behave similarly across nominal formation labels. The model has not failed to separate the classes; it has learned that the classes are not cleanly separable, which is the correct conclusion.

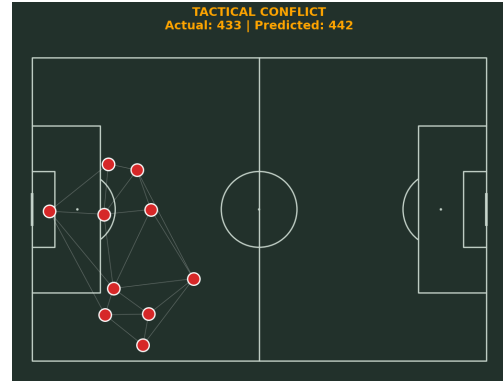


Fig. 7. Visualizing a tactical conflict where the ground truth label (e.g., 4-3-3) is contradicted by the GNN's identification of a 4-4-2 defensive block based on actual spatial coordinates. This illustrates scenarios in which preset labels do not align with observed spatial structure.

C. Ground Truth vs. Game Reality

A significant portion of the exploration involved confronting the discrepancy between official tactical labels and the fluid movements on the pitch. In multiple instances the model predicted a formation that disagreed with the ground truth label, but visual inspection of the spatial graph suggested the model's prediction appeared tactically plausible.

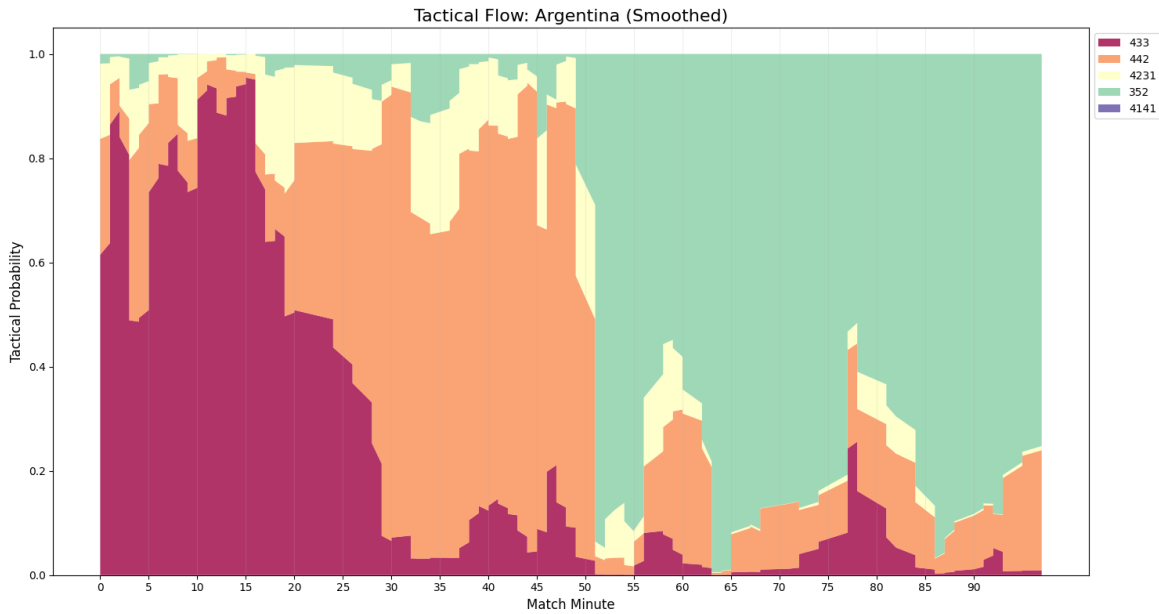


Fig. 8. Refined tactical flow of Argentina. The smoothed probability stackplot illustrates the transition between dominant formations across the full duration of the match. This visualization represents the tactical swings that coincide with coaching decisions and events in the game.

This points to a deeper limitation of supervised formation classification: the ground truth labels are human-annotated and static for long periods, reflecting a coach’s declared intention rather than the team’s actual spatial structure at any given moment. A model that achieves 91% accuracy against these labels is not necessarily correct 91% of the time. In transition moments and tactical phases where the label has not been updated, the model’s spatiotemporal inference may be more temporally precise than the annotation it is being evaluated against. This suggests that future work should move toward unsupervised or self-supervised learning, where the model identifies tactical clusters based entirely on spatial movement rather than conforming to potentially mismatched human labels [14].

D. Match Dynamics: The 2022 World Cup Final

The clearest demonstration of the model’s analytical value came from applying it to the 2022 FIFA World Cup Final between Argentina and France. The smoothed formation probability stackplot (Fig. 8) transforms the model’s frame-by-frame outputs into a legible tactical narrative of the full match.

Argentina opened the match in a 4-3-3, reflecting an aggressive attacking posture consistent with their opening dominance; they led 2-0 by the 36th minute. The model captures a gradual shift toward a 4-4-2 shape through the first half as Argentina consolidated possession and began to manage the game. From the 50th minute onward the model predominantly predicts a 3-5-2, coinciding with Argentina’s transition into a more defensive structure. At the 61st minute, Argentina’s manager substituted an attacking player for a defender, a decision the model reflects as a sustained shift toward a five-defender shape. The absence of a five-defender formation label in the StatsBomb taxonomy likely causes the model to map this structure onto the nearest available label,

but the directional trend in the stackplot correctly captures the tactical intent: Argentina were less aggressive defending their lead.

The spike of 4-4-2 and 4-3-3 probability between the 78th and 84th minutes is the most analytically interesting feature of the plot. France scored from a penalty in the 80th minute and again in the 81st, pulling level at 2-2. The model’s instability during this window is consistent with the set-piece disruption discussed below, but it also reflects genuine tactical chaos: Argentina were conceding rapidly, their defensive structure was breaking down, and the graph was no longer encoding a stable formation state.

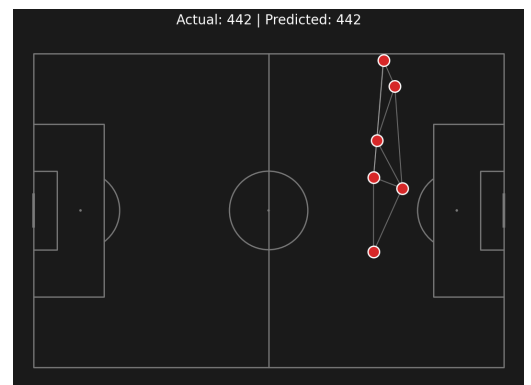


Fig. 9. A spatial graph demonstrating the model’s ability to maintain classification confidence despite missing player data points. A normal soccer team consists of 10 field players and one goalie. Here the model recognizes the midfield line of 4 with 2 strikers in front of them.

E. Set Pieces and the Limits of Spatial Graphs

During corner kicks and free kicks, players from all positions congregate in a high-density zone, collapsing the team’s

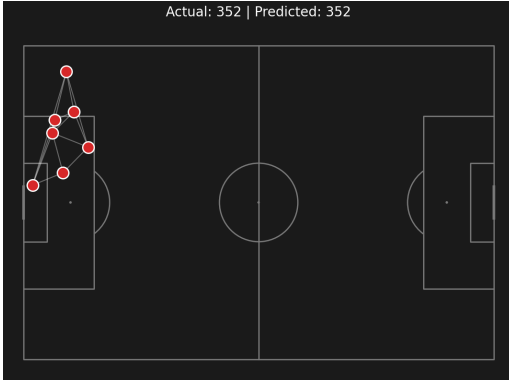


Fig. 10. A visualization of spatial collapse during a set-piece event. This illustrates how high-density scenarios challenge the model’s ability to maintain a standard formation layout.

spatial graph into a structure that bears no resemblance to any formation. The model maintains reasonable accuracy through these events, but the mechanism is not spatial inference, it is temporal memory. The GRU carries the formation state from the frames preceding the set piece through the disruption, effectively bridging the gap until the team returns to open play. The Delaunay triangulation over a dense cluster produces a graph topology that is meaningless as a formation signal; the model survives set pieces because it is remembering rather than reading.

This is a fundamental limitation of the spatial graph representation rather than a failure of the temporal architecture. A context-aware layer that identifies set-piece events and suspends formation inference until open play resumes would address the symptom. The deeper fix would require a formation-agnostic model specifically trained on set-piece structures, ensuring that high-density scenarios are interpreted on their own terms rather than forced into the formation classification framework.

VII. FUTURE DIRECTIONS

A. A Two-Tier Landscape

GNN-based tactical analysis will continue to grow, but not uniformly. The infrastructure required, high-frequency tracking systems, proprietary data pipelines, and the engineering capacity to deploy STGNNs at match speed, is expensive. Elite clubs in the Premier League, Bundesliga, and Champions League are already investing heavily in data analytics [14], [12], and that investment will deepen. For the majority of professional clubs operating on tighter budgets, the gap between what the data architecture can do and what they can afford to run will widen before it narrows. The result is a two-tier landscape: a small number of well-resourced clubs with genuinely predictive relational models, and everyone else still working from event-based statistics.

B. Real-Time Tactical Analysis

The most significant near-term development is the move from post-match analysis to real-time deployment. The

STGNN framework is already theoretically capable of computing expected threat changes across player rotations and formation matchups on a live graph, the architecture does not change, only the latency requirement does. As hardware and inference pipelines mature, the natural application is in-match tactical feedback: a system that tracks how a team’s xT distribution shifts as players rotate positions, flags when a formation matchup is being structurally exploited, and surfaces that information to a coach’s staff in real time.

This is also where the unified multi-task model described becomes most valuable. A model that simultaneously tracks formation state at the graph level and individual player contributions at the node level would give coaching staff both lenses at once: not just that the team’s shape is degrading, but which specific player’s positional drift is causing it.

C. Recruitment: xT Alongside xG , Not Instead

Expected goals will not be displaced. It is interpretable, widely understood, and good at what it measures. What GoalNet-style frameworks offer is not a replacement but a complement [8]: a way to surface the defensive midfielder who never appears in the xG log but whose connective centrality is holding the team’s structure together. In recruitment, where clubs are specifically looking for undervalued players, players whose contributions are real but invisible to the metrics rivals are using, this is exactly the kind of edge that justifies the investment. The most likely near-term outcome is that xT and centrality-based credit systems become standard tools in elite recruitment pipelines alongside xG , not instead of it.

D. Gender-Specific Modeling as Standard Practice

This project itself is a case in point. The model developed in Section VI was trained exclusively on a single men’s international match. Whether its learned spatial signatures, the connectivity patterns it associates with a 4-3-3 or a 3-5-2, would transfer meaningfully to women’s club or international data is an open question. The differences may be subtle or significant, but they cannot be assumed away. As the field expands into women’s leagues, the relational patterns specific to that game need to be studied on their own terms, with dedicated datasets and models built from the ground up rather than adapted from men’s architectures. Bekkers and Sahasrabudhe’s findings establish that the difference exists and matters. The work of characterizing exactly what that difference is has barely begun [5].

VIII. CONCLUSION

Soccer is a relational system, and understanding it computationally requires a model that treats relationships as first-class data. Graph Neural Networks provide that model. By representing players as nodes, spatial and tactical relationships as edges, and match sequences as time-evolving graphs, the STGNN framework encodes the structure of soccer in a way that traditional metrics and grid-based models cannot.

The aggregation function is not a neutral implementation detail. Sum aggregation preserves the collective connectivity

structure that encodes a formation; attention aggregation learns to identify the specific interactions that drive individual player contributions. Selecting the wrong function for a given task means the model is structurally incapable of representing the information the task requires. The temporal window length τ introduces a parallel design tension between sensitivity to rapid transitions and stability over longer structural patterns, and oversmoothing places an opposing constraint on network depth.

Formation recognition and tactical analysis of the players illustrate how the same framework operates on different scales. The Explorations experiment grounds these principles in a working system: 91.14% validation accuracy on a four-class formation problem, with per-formation metrics that reveal the model's internal logic as much as its performance. The geometric distinctiveness of 3-5-2 makes it the most classifiable formation; the overlap between all other clusters in the latent space reflects the continuous and fluid nature of tactical identity in professional soccer. The model's uncertainty mirrors the game's own ambiguity, and its disagreements with ground truth labels are sometimes more tactically accurate than the labels themselves.

Open questions around adaptive aggregation, temporal window design, multi-task learning, gender-specific modeling, and data access will determine whether GNN-based soccer analytics can continue to grow into deployment at the professional level. The framework is solid. The constraints are practical.

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APPENDIX

Synthesis of Coursework

The development of this STGNN served as a practical synthesis of the core pillars of my Computer Science education. The interdisciplinary nature of soccer analytics required the application of principles from several courses:

- **Statistics (Math 124) and Data 162/272:** These courses provided the critical foundation for handling class imbalance. The decision to implement specific weighting for the minority 3-5-2 formation was a direct application of statistical sampling and bias-correction techniques. This ensured that the model's 91.14% accuracy was a reflection of true pattern recognition rather than simple majority-class bias.
- **Machine Learning (CSCI 332):** The core architecture—integrating Graph Convolutional Networks (GCN) with Gated Recurrent Units (GRU)—was an exercise in model selection and hyperparameter optimization. Understanding the trade-offs between spatial connectivity and temporal decay was essential to successfully classifying complex formations.
- **Database Systems (in Sevilla, Spain):** Managing the StatsBomb 360 dataset required efficient preprocessing. The logic used to join high-frequency tracking frames with event-level data relied on the relational and indexing principles mastered in my database coursework, ensuring the data pipeline was robust before the graph construction phase.
- **Algorithms and Concurrency (CSCI 338):** Constructing the graph required implementing Delaunay Triangulation to define spatial edges. Processing these graphs in temporal sequences ($\tau = 15$) required an understanding of algorithmic complexity and memory management to ensure the pipeline operated efficiently during the multi-match inference phase.

Personal Reflection

Since changing my major to Computer Science, I have been deeply interested in how computational models can represent the physical world. This project was my first opportunity to move beyond classroom theory and attempt a full-scale deep learning implementation independently.

To finally translate raw positional data into a model that can interpret a game has been an extremely fun and rewarding experience. The process of debugging the "tactical noise" in the Argentina timeline and seeing a Gaussian filter transform it into a coherent narrative was a highlight of the exploration. This project has affirmed my passion for Data Science, proving that the most complex problems—like the fluid motion of a soccer match—can be understood through the lens of well-designed algorithms.

Thank you for a great semester!

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